

Build "LaunchPad" – A Student Project Showcase & Innovation Platform

Project Overview

Develop a modern web application called **LaunchPad**, a platform where college students can showcase their academic and personal projects, build a professional portfolio, participate in innovation contests, receive feedback from peers, and gain recognition through rankings and achievements.

The platform should feel like a combination of **GitHub** + **Product Hunt** + **LinkedIn**, but designed specifically for students and campus communities.

The objective is to create an ecosystem where students can publicly present their work, connect with other creators, and demonstrate their practical skills to recruiters, mentors, and fellow students.

Problem Statement

Most students complete impressive academic or personal projects during their education, but these projects often remain stored on personal computers or are submitted only for internal college evaluation.

There is no centralized platform dedicated to helping students:

- Present their projects professionally
- Build a technical portfolio
- Receive constructive feedback
- Discover projects created by other students
- Participate in project competitions
- Earn recognition for their work
- Showcase their skills to potential employers

LaunchPad solves this problem by providing a dedicated platform where innovation is visible, collaborative, and rewarding.

Objectives

The platform should allow students to:

- Create a professional profile
- Upload and manage projects
- Showcase technical skills
- Share project demos and repositories
- Receive likes, comments, and bookmarks
- Participate in contests
- Earn badges and achievements
- Build credibility through community engagement

Administrators should be able to manage the platform, moderate content, organize contests, and feature outstanding projects.

User Roles

Guest

Guests can:

- Browse public projects
- Search projects
- Explore categories
- View public profiles
- View featured projects
- View leaderboard

Guests cannot interact with projects until they create an account.

Registered Student

Students can:

- Register
- Login
- Manage profile
- Upload projects
- Edit projects

- Delete projects
 - Upload screenshots
 - Add live demo links
 - Add repository links
 - Add documentation
 - Follow other students
 - Like projects
 - Bookmark projects
 - Comment on projects
 - Receive notifications
 - Join contests
 - Earn badges
 - View personal analytics
 - Appear on the leaderboard
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Administrator

Administrators can:

- Manage users
 - Manage projects
 - Approve or remove inappropriate content
 - Manage categories
 - Manage contests
 - Feature projects
 - Create badges
 - View analytics
 - Manage reports
 - Configure platform settings
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User Workflow

Registration

A new student creates an account using:

- Username
- Email
- Password

After successful registration:

- Email verification is required.
 - The account becomes active after verification.
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Login

After authentication, students are redirected to their personalized dashboard.

Dashboard

The dashboard acts as the student's workspace.

It displays:

- Recent notifications
- Profile completion
- Total projects
- Followers
- Following
- Total views
- Likes received
- Comments received
- Contest participation
- Earned badges
- Activity summary

Profile

Every student has a public profile.

The profile includes:

- Avatar
- Cover image
- Display name
- Username
- College
- Education
- Skills
- Bio
- Portfolio URL
- Social links
- Achievements
- Followers
- Following
- Uploaded projects
- Earned badges

Visitors can follow the student and explore all published projects.

Project Management

Students can create unlimited projects.

Each project contains:

- Project title
- Short description
- Detailed description
- Category
- Technologies used
- Screenshots
- Live demo link
- Source code repository
- Documentation
- Status (Completed / Ongoing / Prototype)
- Visibility (Public / Private)
- Tags

Students can:

- Create
- Edit
- Delete
- Publish
- Unpublish

Project Discovery

Visitors can discover projects using:

- Search
- Categories
- Tags
- Trending
- Featured
- Latest
- Most Viewed
- Most Liked

Filtering options should allow users to quickly find relevant projects.

Social Features

Students should be able to:

- Like projects
- Bookmark projects
- Organize bookmarks into collections
- Comment on projects
- Follow creators
- View followers
- View following

Interactions should encourage community engagement.

Notifications

The platform should notify users when:

- Someone follows them
- Someone likes a project
- Someone comments
- Contest results are announced
- A badge is awarded
- A featured project is selected

Users should be able to:

- View notifications
 - Mark notifications as read
 - Clear notifications
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Bookmarks & Collections

Students can save interesting projects.

Bookmarks can be organized into custom collections such as:

- AI Inspiration
- Final Year Ideas
- UI Designs
- Machine Learning
- Web Development

Collections help students organize inspiration.

Contest System

Administrators can organize project contests.

Each contest contains:

- Title
- Description
- Rules
- Registration deadline
- Submission deadline
- Maximum participants
- Prize information
- Status

Students can:

- Register
 - Submit projects
 - Update submissions before the deadline
 - View contest results
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Contest Evaluation

Administrators can review submissions.

Winning projects receive:

- Certificates

- Badges
 - Leaderboard points
 - Recognition on the homepage
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Leaderboard

Students earn points through meaningful participation.

Points may come from:

- Published projects
- Likes
- Comments
- Followers
- Contest achievements
- Featured projects
- Badges

Leaderboards should support:

- Weekly rankings
 - Monthly rankings
 - Overall rankings
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Badges & Achievements

Students can unlock achievements such as:

- First Project
- Top Creator
- Contest Winner
- Community Favorite
- Rising Innovator
- Most Viewed
- Most Liked

Achievements motivate continuous contribution.

Search System

Search should support:

- Student names
- Project titles
- Categories
- Technologies
- Skills
- Colleges
- Tags

Search should provide relevant and quick results.

Analytics

Students should have access to personal analytics, including:

- Total project views
- Unique visitors
- Likes
- Comments
- Followers growth
- Profile visits
- Most viewed project
- Engagement trends

Administrators should have platform-wide analytics such as:

- Daily active users
- New registrations
- Most popular categories

- Contest participation
 - Featured project performance
 - User growth
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Homepage

The homepage should include:

- Hero section
- Featured projects
- Trending projects
- Categories
- Contest highlights
- Student testimonials
- Platform statistics
- Call-to-action sections

The homepage should encourage visitors to explore and register.

Admin Dashboard

The administrator dashboard should provide complete control over the platform, including:

- User management
 - Project moderation
 - Category management
 - Contest management
 - Badge management
 - Featured project selection
 - Reports and analytics
 - Platform configuration
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Security Requirements

The platform should include:

- Authentication
 - Email verification
 - Password reset
 - Session management
 - CSRF protection
 - Input validation
 - Role-based authorization
 - Secure file uploads
 - Rate limiting for sensitive actions
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Expected Outcome

LaunchPad should feel like a professional SaaS platform built for the student ecosystem rather than a typical academic CRUD project.

The final application should encourage students to continuously build, publish, improve, and showcase their work while creating an active community around innovation, collaboration, and technical growth. It should be polished enough to be used as a real portfolio platform and compelling enough that a recruiter, mentor, or college faculty member could browse student projects, discover talent, and evaluate practical skills through the platform.